Simple Mathematical Intuition of Module 14

* Sigmoid function

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

sigma of z

z = the linear score (zeta)

$\sigma(z) = \sigma(0) = 0.5$

$\sigma(z) = \sigma(z) \to 1$ as $z \to +\infty$

$\sigma(z) \to 0$ as $z \to -\infty$

* Linear score:

$$z = w^T x + b$$

* Probability Estimation:

$$p = \sigma(z) = \frac{1}{1+e^{-z}}$$

prediction rule: $\hat{y} = \begin{cases} 1 & \text{if } p \geq 0.5 \\ 0 & \text{if } p < 0.5 \end{cases}$

* Odds

How likely it is to happen vs how likely it is to not happen

The ratio is called odds:

$$\text{odds} = \frac{p}{1-p}$$

if, $p = 0.5$ $\quad$ $\text{odd} = \frac{0.5}{1-0.5} = \frac{0.5}{0.5} = 1$

if, $p = 0.8$ $\quad$ $\text{odd} = \frac{0.8}{1-0.8} = \frac{0.8}{0.2} = 4$

$\frac{0.2}{\phantom{0}} = 0.25 = \frac{1}{4}$th

if, $p = 0.8$ $\quad$ odd $= \frac{0.8}{1-0.8} = 0.2$

if, $p = 0.2$ $\quad$ odd $= \frac{0.2}{1-0.2} = \frac{0.2}{0.8} = 0.25 = \frac{1}{4}$th

* Log Odd:

$$\log\left(\frac{p}{1-p}\right)$$

logit function

$$\log\left(\frac{p}{1-p}\right) = w^T x + b$$

Linear model

Transformation of probability

odd $(0 \quad \infty)$

$(-\infty, +\infty)$

$p = 0.7$

$\frac{0.7}{0.3} = 2.33$

$\log(2.33)$

$= 0.847$

* Decision Boundary:

$p = 0.5$

$$\frac{p}{1-p} = 1$$

$\log(1) = 0$

Thus, $w^T x + b = 0 \rightarrow$ The decision boundary.

* Example:

$w = 2, \quad b = -6$

$\Rightarrow$

$2x - 6 = 0$

$x = 3$

predicts class 1

" 0

$x = 3$

if, $x > 3$ : model predicts class 1

if $x < 3$ : " " " 0

if $x = 3$ : probability is exactly 0.5

* Binary cross Entropy (Log loss)

cost function:

$$J(w,b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log(p^{(i)}) + (1-y^{(i)})\log(1-p^{(i)})\right]$$

$\downarrow$

cost value

$y^{(i)}$ = true label

$p^{(i)}$ = predicted probability

* Gradient Descent variables

Learing rate : $\alpha$

Gradient updates:

$$w: w - \alpha\frac{\partial J}{\partial w}$$

$$b: b - \alpha\frac{\partial J}{\partial b}$$

$\leftarrow 1 \rightarrow 1.5$

0.45

Gradients:

$$\frac{\partial J}{\partial w} = \frac{1}{m}\sum_{i=1}^{m}(p^{(i)} - y^{(i)}x^{(i)})$$

$$\frac{\partial J}{\partial b} = \frac{1}{m}\sum_{i=1}^{m}(p^{(i)} - y^{(i)})$$

Model: $w = 0.8$ , $b = -3$

Student studies: $x = 5$ hours

Step 1: Compute $z$

$$z = wx + b = 0.8 \times 5 - 3$$
$$= 1$$

Step 2: Apply sigmoid

$$p = \sigma(1) = \frac{1}{1+e^{-1}}$$
$$= 0.731$$

Step 3: Class prediction

$$p = 0.731 \geq 0.5 \Rightarrow \hat{y} = 1$$

Step 4: Decision boundary

$$wx + b = 0$$
$$0.8x - 3 = 0$$
$$x = \frac{3}{0.8} = 3.75$$

$S.H \geq 3.75 \rightarrow$ pass

$S.H < 3.75 \rightarrow$ fail

* Example of Gradient Descent:

$w = 0.5$ $b = -1$

Training example:

$$x^{(1)} = 2, \; y^{(1)} = 1$$

We need to compute: $z, p, \frac{\partial J}{\partial w}, \frac{\partial J}{\partial b}$

Step 1: Compute Linear score

$$= 0.5 \times 2 - 1$$

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$$z = wx^{(1)} + b = (0.5) \times 2 - 1$$
$$= 1 - 1 = 0$$

Step 2:

$$p = \sigma(0) = 0.5$$

Step 3: Compute gradient of loss

$$\frac{\partial J}{\partial w} = (p - y)x$$

$$\frac{\partial J}{\partial b} = (p - y)$$

$$p - y = 0.5 - 1 = -0.5$$

$$\frac{\partial J}{\partial w} = (p - y) \cdot x = -0.5 \times 2 = -1$$

$$\frac{\partial J}{\partial b} = -0.5$$

Step 4: One gradient descent update:

$$\alpha = 0.1$$

Update rules:

$$w: w - \alpha \frac{\partial J}{\partial w}$$

$$b: b - \alpha \frac{\partial J}{\partial b}$$

$$w_{new} = 0.5 - (0.1)(-1) = 0.6$$

$$b_{new} = -1 - (0.1)(-0.5) = -0.95$$

$$z_{new} = (0.6)(2) - 0.95 = 1.2 - 0.95 = 0.25$$

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$$P_{new} = \sigma(0.25) = \underline{0.562}$$